Chemistry 3A – Spring 2026 **Aleyda Hernandez Zamorano 1065482** RETAKE VERSION
Sections 19320-19321
Midterm Examination #3 8 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law

2. Convert a pressure of 1.50 atm to millimeters of mercury (mmHg)
(a) 507 mmHg (b) 760 mmHg (c) 1,140 mmHg (d) 1,500 mmHg

3. In a redox reaction, the species that ***loses electrons*** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) unchanged

4. A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., A + B → AB) is what type of reaction?
(a) decomposition (b) single displacement (SD)
(c) double displacement (DD) (d) combination (synthesis) reaction

5. What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M

6. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M
$M_1V_1 = M_2V_2$
$10.0 \times 2.5 \div 500 = 0.50\ M$

7. A solution is prepared by mixing 25.0 g of solute with 75.0 g of solvent. What is the percent by mass (m/m%) of the solute?
(a) 25.0% (b) 33.3% (c) 75.0% (d) 100.0%
$\frac{25.0}{75.0} \times 100 = 33.333$

8. Calculate the **pressure** of n = 2.00 mol of gas at T = 273 K, V = 44.8 L. (R = 0.0821 L·atm/[mol·K])
(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm

9. Convert a **pressure** of 760 mmHg to atmospheres (atm).
(a) 0.500 atm (b) 1.00 atm (c) 7.60 atm (d) 76.0 atm

10. Which of the following is a ***combustion*** reaction?
(a) $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ (b) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(c) $HCl + NaOH \rightarrow NaCl + H_2O$ (d) $CaCO_3 \rightarrow CaO + CO_2$

11. Which of the following is the correct oxidation number of ***hydrogen*** in H_2O molecule?
(a) 0 (b) -1 (c) +1 (d) +2

12. A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
(a) non-electrolyte (b) weak electrolyte (c) strong electrolyte (d) nonpolar molecule

13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2?
(a) -1 (b) 0 (c) +1 (d) +7

14. The strongest type of ***intermolecular force*** that exists between two molecules of H_2O (water) is:
(a) dispersion forces (London) (b) dipole-dipole forces
(c) ion-dipole forces (d) hydrogen bonding

15. How many grams of KOH are required to prepare 1.00 L of a 0.500 M solution?
(molar mass of KOH = 56.11 g/mol)
(a) 5.61 g (b) 28.1 g (c) 56.1 g (d) 112.2 g

guessed
$0.500 = 1000 \times 56.11$
$= 0.028055$
$\approx 28.1\,g$

16. Which is a TRUE statement of the **Kinetic Molecular Theory** for an *ideal gas*?
(a) gas molecules exert significant attractive forces on each other
(b) the molecules that make up the gas phase must interact with their liquid phase
(c) gas collisions are completely elastic (energy is not lost in collision)
(d) the volume (V) of the gas is inversely proportional to the number of moles (n)

17. Which of the following compounds is ***insoluble*** (forms precipitate) in water?
(a) KNO_3 (b) Na_2SO_4 (c) AgCl (d) LiCl

18. A solution that has not yet reached its upper limit of dissolved solute is described as:
(a) unsaturated (b) supersaturated (c) saturated (d) concentrated

19. A gas has initial pressure of 2.00 atm and volume of 5.00 L. If the volume is compressed to 2.50 L at constant temperature, what is the new pressure?
(a) 1.00 atm (b) 4.00 atm (c) 5.00 atm (d) 10.0 atm

20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
(a) S, Cl, Br (b) Cl, Br, I (c) N, O, F (d) Na, K, Li

21. In a sugar-water solution, the sugar is the ***solute*** and the water is the ***solvent***. The solvent is defined as the component present in:
(a) the smallest amount (b) the solid phase
(c) the greatest amount (d) the ionic form

22. The concentration unit of ***molarity*** (***M***) is correctly expressed in which of the following units?
(a) mol/kg (b) mol/L (c) gl/L (d) kg/mol

23. Which of the following is a spectator ion in the reaction (hint: compare complete ionic to net ionic):
$NaCl\,(aq) + AgNO_3\,(aq) \rightarrow AgCl\,(s) + NaNO_3\,(aq)$?
(a) Na^+ (b) Ag^+ (c) Cl^- (d) AgCl (S) becomes solid.

24. Which statement t best describes the principle of "like dissolves like"?
(a) polar solutes dissolve in polar solvents (b) nonpolar solutes dissolve in nonpolar solvents
(c) ionic solutes dissolve in polar solvents (d) all of choices (a) to (c)

25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
(a) ***(s)*** (b) ***(l)*** (c) ***(g)*** (d) ***(aq)***

Precipitate = Insoluble Compoun (S) Solid

Circle either TRUE or FALSE whether the statement is true or false.

26. A solution that conducts electricity must contain ions.
(True) False

27. The reaction HCl (*aq*) + NaOH (*aq*) → H_2O (*l*) + NaCl (*aq*) is an acid-base neutralization reaction type.
(True) False

28. The balancing of a chemical reaction equation involves changing the subscripts of a chemical compound formula, like changing **$CuCl_2$** to **CuCl**
True (False)

29. The oxidation number of a pure element is always zero.
True (False) Oxidation looses an electron

30. In a chemical reaction equation, compounds to the ***right*** of the **reaction arrow** (yield symbol) are called ***reactants***. Reactants → Products
True (False)

31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.
(True) False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.
(True) False

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate.** = insoluble compound
(True) False

34. At STP, the temperature is 0°C and pressure is 1 atm.
(True) False

35. The symbol ***n*** in the Ideal Gas Law equation (*PV=nRT*) is known as the ***universal gas constant***.
True (False) n = number of moles
R = gas constant

Chemistry 3A – Spring 2026 **Leah Anchondo** **1012027** RETAKE VERSION

Sections 19320-19321

Midterm Examination #3 8 May 2026

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6. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)? 2.5M · 10.0mL = 25
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M 25 / 50.0mL = 0.50

7. A solution is prepared by mixing 25.0 g of solute with 75.0 g of solvent. What is the percent by mass (m/m%) of the solute?
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Chemical compounds Left 2 arrow are reactant
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True False

P = Pressure
V = Volume
T = Temp

Chemistry 3A – Spring 2026 **Noah Cruz** **1044628** RETAKE VERSION
Sections 19320-19321
Midterm Examination #3 8 May 2026

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(a) **(*s*)** (b) **(*l*)** (c) **(*g*)** (d) **(*aq*)**

Circle either TRUE or FALSE whether the statement is true or false.

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True False

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True False

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True False

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True False

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True False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.
True False

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True False

34. At STP, the temperature is 0°C and pressure is 1 atm.
True False

35. The symbol ***n*** in the Ideal Gas Law equation (*PV=nRT*) is known as the ***universal gas constant***.
True False

Chemistry 3A – Spring 2026 **Cortney Lewis 0893445** RETAKE VERSION

Sections 19320-19321

Midterm Examination #1 8 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which element belongs to the **alkali metals** group?
(a) calcium (b) sodium (c) magnesium (d) chlorine

2. Which subatomic particle has a mass of approximately **1 amu** and a positive charge?
(a) electron (b) proton (c) neutron (d) none of the above

3. Which of the following is a **polyatomic ion**?
(a) Cl^- (b) Na^+ (c) SO_4^{2-} (d) O^{2-}

4. Convert 25.0 °C into Kelvin.
(a) 0 K (b) 25 K (c) 273 K (d) 298 K

5. A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density?
(a) 0.200 g/mL (b) 5.00 g/mL (c) 31.3 g/mL (d) 0.050 g/mL

6. Which of the following is the correct name for $Ca(NO_3)_2$?
(a) cobalt fluoride (b) calcium nitroxide (c) calcium nitrate (d) calcium oxide

7. Which of the following terms describes the distance between two wave crests?
(a) wave speed (b) frequency (c) wavelength (d) amplitude

8. Which of the following is the correct atomic notation for carbon-14?
(a) ${}^{12}_{6}C$ (b) ${}^{14}_{6}C$ (c) ${}^{6}_{14}C$ (d) ${}^{60}_{27}Co$

9. Convert 4.56×10^{-3} into standard notation
(a) 0.00456 (b) 4560 (c) 0.000456 (d) 4,560

10. Which of the following is the electron configuration for hydrogen?
(a) $1s^1$ (b) $1s^2$ (c) $1s^2 2s^1$ (d) $1s^2 2p^1$

11. Which of the following is TRUE about element bromine (Br)?
(a) It is a member of the *d*-block of elements
(b) It is a Group 1 element
(c) It is a non-metal
(d) none of the above

12. What term describes an electron normally in $n = 1$ shell acquiring energy to be in $n=2$ shell?
(a) emission (b) excited state (d) ground state (d) absorption

13. Express 0.0000721 in scientific notation with correct significant figures.
(a) 7.21×10^{-4} (b) 7.21×10^{-5} (c) 7.21×10^{-6} (d) 7.21×10^{-7}

14. When you complete the operation on the expression $1.450 \times 3.01 / 10.025$, how many significant digits will the final result have? ('/' = division operation)
(a) 2 (b) 3 (c) 4 (d) 5

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?
(a) law (b) hypothesis (c) theory (d) experimentation

16. Which of these is a polyatomic ion?
(a) Ne (b) CO_3^{2-} (c) Br^- (d) Na^+

17. Which of these is the ground-state electron configuration for the **carbon (C)** atom?
(a) $1s^2 2s^2 2p^6$ (b) $1s^2 2s^2 2p^2$ (c) $1s^2 2s^2 2p^4$ (d) $1s^2 2s^2$

18. How many quantum numbers are required to uniquely define the state of an electron in an atom?
(a) 1 (b) 2 (c) 3 (d) 4

19. Perform the operation: 3.456 + 12.1 + 0.0045. Report with correct decimal places.
(a) 15.5605 (b) 15.56 (c) 15.6 (d) 15.561

20. The label on the bottle has the formula $FeCl_3$. What is its name using the Stock system?
(a) iron chloride (b) iron(II) chloride (c) iron(III) chloride (d) iron monochloride

21. One of the quantum numbers of an electron is related to the shape of the electron's orbital and is represented by a letter in this set: {*s*, *p*, *d*, *f*}. Which quantum number is this?
(a) *n* (b) *l* (c) m_s (d) Z

22. Which of these subatomic particles is **NOT** found in the nucleus of the atom?
(a) proton (b) neutron (c) electron (d) none of the choices (a) to (c)

23. What is a TRUE statement regarding an element in the Periodic Table?
(a) the number of neutrons and protons for every element must be equal
(b) the atomic number (Z value) indicates the number of protons it has
(c) the mass number (A value) indicates the number of electrons and protons it has
(d) the isotope of the element is determined by the number of electrons

24. What are the Group 2 elements called?
(a) Noble gases (b) Alkali metals (c) Alkaline Earth metals (d) Halogens

25. An isotope of an element has atomic number (Z) = 15 and mass number (A) of 31. How many neutrons are in the nucleus?
(a) 15 (b) 16 (c) 31 (d) 46

Circle either TRUE or FALSE whether the statement is true or false.

26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.
True (False)

27. Ionic compounds with monatomic elements are typically formed between metals and nonmetals.
(True) False

28. The speed/velocity of a electromagnetic wave can be calculated by dividing the frequency by the wavelength: $c = \nu / \lambda$
True (False) $\frac{c}{\lambda \times \nu}$

29. The **p-block** of elements in the Periodic Table starts in Period 1.
True (False)

30. The **Pauli Exclusion Principle** states that an orbital can have only two electrons which are paired up with opposite spins
(True) False

31. According to the rules for significant figures, **conversion factors** (like 1 m = 100 cm) are considered to have infinite significant digits and do not affect the precision of a calculation.
True (False)

32. The atomic number (*Z*) represents the number of protons in an atom.
(True) False

33. The only subatomic particles allowed in the nucleus of the atom are **protons**.
True (False)

34. Noble gases have complete valence shells, which is why they do not form compounds with other elements.
(True) False

35. The law of conservation of mass states that matter **CAN** be created or destroyed in chemical reactions.
True (False)

Chemistry 3A – Spring 2026 **Tayler K. Rios 0934356** RETAKE VERSION

Sections 19320-19321

Midterm Examination #1 8 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which element belongs to the **alkali metals** group?
(a) calcium (b) sodium (c) magnesium (d) chlorine

2. Which subatomic particle has a mass of approximately **1 amu** and a positive charge?
(a) electron (b) proton (c) neutron (d) none of the above

3. Which of the following is a **polyatomic ion**?
(a) Cl^- (b) Na^+ (c) SO_4^{2-} (d) O^{2-}

4. Convert 25.0 °C into Kelvin.
(a) 0 K (b) 25 K (c) 273 K (d) 298 K

5. A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density?
(a) 0.200 g/mL (b) 5.00 g/mL (c) 31.3 g/mL (d) 0.050 g/mL

6. Which of the following is the correct name for $Ca(NO_3)_2$?
(a) cobalt fluoride (b) calcium nitroxide (c) calcium nitrate (d) calcium oxide

7. Which of the following terms describes the distance between two wave crests?
(a) wave speed (b) frequency (c) wavelength (d) amplitude

8. Which of the following is the correct atomic notation for carbon-14?
(a) $^{12}_{6}C$ (b) $^{14}_{6}C$ (c) $^{6}_{14}C$ (d) $^{60}_{27}Co$

9. Convert 4.56×10^{-3} into standard notation
(a) 0.00456 (b) 4560 (c) 0.000456 (d) 4,560

10. Which of the following is the electron configuration for hydrogen?
(a) $1s^1$ (b) $1s^2$ (c) $1s^2 2s^1$ (d) $1s^2 2p^1$

11. Which of the following is TRUE about element bromine (Br)?
(a) It is a member of the *d*-block of elements
(b) It is a Group 1 element
(c) It is a non-metal
(d) none of the above

12. What term describes an electron normally in $n = 1$ shell acquiring energy to be in $n=2$ shell?
(a) emission (b) excited state (d) ground state (d) absorption

13. Express 0.0000721 in scientific notation with correct significant figures.
(a) 7.21×10^{-4} (b) 7.21×10^{-5} (c) 7.21×10^{-6} (d) 7.21×10^{-7}

14. When you complete the operation on the expression 1.450 × 3.01 / 10.025, how many significant digits will the final result have? ('/' = division operation)
(a) 2 (b) 3 (c) 4 (d) 5

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?
(a) law (b) hypothesis (c) theory (d) experimentation

16. Which of these is a polyatomic ion?
(a) Ne (b) CO_3^{2-} (c) Br^- (d) Na^+

17. Which of these is the ground-state electron configuration for the **carbon (C)** atom?
(a) $1s^2 2s^2 2p^6$ (b) $1s^2 2s^2 2p^2$ (c) $1s^2 2s^2 2p^4$ (d) $1s^2 2s^2$

18. How many quantum numbers are required to uniquely define the state of an electron in an atom?
(a) 1 (b) 2 (c) 3 (d) 4

19. Perform the operation: 3.456 + 12.1 + 0.0045. Report with correct decimal places.
(a) 15.5605 (b) 15.56 (c) 15.6 (d) 15.561

20. The label on the bottle has the formula $FeCl_3$. What is its name using the Stock system?
(a) iron chloride (b) iron(II) chloride (c) iron(III) chloride (d) iron monochloride

21. One of the quantum numbers of an electron is related to the shape of the electron's orbital and is represented by a letter in this set: {*s*, *p*, *d*, *f*}. Which quantum number is this?
(a) *n* (b) *l* (c) m_s (d) Z

22. Which of these subatomic particles is **NOT** found in the nucleus of the atom?
(a) proton (b) neutron (c) electron (d) none of the choices (a) to (c)

23. What is a TRUE statement regarding an element in the Periodic Table?
(a) the number of neutrons and protons for every element must be equal
(b) the atomic number (Z value) indicates the number of protons it has
(c) the mass number (A value) indicates the number of electrons and protons it has
(d) the isotope of the element is determined by the number of electrons neutrons

24. What are the Group 2 elements called?
(a) Noble gases (b) Alkali metals (c) Alkaline Earth metals (d) Halogens

25. An isotope of an element has atomic number (Z) = 15 and mass number (A) of 31. How many neutrons are in the nucleus?
(a) 15 (b) 16 (c) 31 (d) 46

Circle either TRUE or FALSE whether the statement is true or false.

26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.
True False

27. Ionic compounds with monatomic elements are typically formed between metals and nonmetals.
True False

28. The speed/velocity of a electromagnetic wave can be calculated by dividing the frequency by the wavelength: $c = \nu / \lambda$
True False

29. The **p-block** of elements in the Periodic Table starts in Period 1.
True False

30. The **Pauli Exclusion Principle** states that an orbital can have only two electrons which are paired up with opposite spins
True False

31. According to the rules for significant figures, **conversion factors** (like 1 m = 100 cm) are considered to have infinite significant digits and do not affect the precision of a calculation.
True False

32. The atomic number (*Z*) represents the number of protons in an atom.
True False

33. The only subatomic particles allowed in the nucleus of the atom are **protons**.
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34. Noble gases have complete valence shells, which is why they do not form compounds with other elements.
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Chemistry 3A – Spring 2026 **Ishmael Lopez 1012110** RETAKE VERSION

Sections 19320-19321

Midterm Examination #1 8 May 2026

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Chemistry 3A – Spring 2026 **Andrew Esparza 0897993** RETAKE VERSION

Sections 19320-19321

Midterm Examination #1 8 May 2026

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Chemistry 3A – Spring 2026 **Ia Ong Lee** **0431206** RETAKE VERSION
Sections 19320-19321
Midterm Examination #3 8 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law

2. Convert a pressure of 1.50 atm to millimeters of mercury (mmHg)
(a) 507 mmHg (b) 760 mmHg (c) 1,140 mmHg (d) 1,500 mmHg

3. In a redox reaction, the species that ***loses electrons*** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) unchanged

4. A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., A + B → AB) is what type of reaction?
(a) decomposition (b) single displacement (SD)
(c) double displacement (DD) (d) combination (synthesis) reaction

5. What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M

6. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M

7. A solution is prepared by mixing 25.0 g of solute with 75.0 g of solvent. What is the percent by mass (m/m%) of the solute?
(a) 25.0% (b) 33.3% (c) 75.0% (d) 100.0%

8. Calculate the **pressure** of n = 2.00 mol of gas at T = 273 K, V = 44.8 L. (R = 0.0821 L·atm/[mol·K])
(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm

9. Convert a **pressure** of 760 mmHg to atmospheres (atm).
(a) 0.500 atm (b) 1.00 atm (c) 7.60 atm (d) 76.0 atm

10. Which of the following is a ***combustion*** reaction?
(a) $CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$ (b) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(c) $HCl + NaOH \rightarrow NaCl + H_2O$ (d) $CaCO_3 \rightarrow CaO + CO_2$

11. Which of the following is the correct oxidation number of ***hydrogen*** in H_2O molecule?
(a) 0 (b) -1 (c) +1 (d) +2

12. A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
(a) non-electrolyte (b) weak electrolyte (c) strong electrolyte (d) nonpolar molecule

13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2?
(a) -1 (b) 0 (c) +1 (d) +7

14. The strongest type of ***intermolecular force*** that exists between two molecules of H_2O (water) is:
(a) dispersion forces (London) (b) dipole-dipole forces
(c) ion-dipole forces (d) hydrogen bonding

15. How many grams of KOH are required to prepare 1.00 L of a 0.500 M solution? (molar mass of KOH = 56.11 g/mol)
(a) 5.61 g (b) 28.1 g (c) 56.1 g (d) 112.2 g

16. Which is a TRUE statement of the **Kinetic Molecular Theory** for an *ideal gas*?
(a) gas molecules exert significant attractive forces on each other
(b) the molecules that make up the gas phase must interact with their liquid phase
(c) gas collisions are completely elastic (energy is not lost in collision)
(d) the volume (*V*) of the gas is inversely proportional to the number of moles (*n*)

17. Which of the following compounds is ***insoluble*** (forms precipitate) in water?
(a) KNO_3 (b) Na_2SO_4 (c) AgCl (d) LiCl

18. A solution that has not yet reached its upper limit of dissolved solute is described as:
(a) unsaturated (b) supersaturated (c) saturated (d) concentrated

19. A gas has initial pressure of 2.00 atm and volume of 5.00 L. If the volume is compressed to 2.50 L at constant temperature, what is the new pressure?
(a) 1.00 atm (b) 4.00 atm (c) 5.00 atm (d) 10.0 atm

20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
(a) S, Cl, Br (b) Cl, Br, I (c) N, O, F (d) Na, K, Li

21. In a sugar-water solution, the sugar is the ***solute*** and the water is the ***solvent***. The solvent is defined as the component present in:
(a) the smallest amount (b) the solid phase
(c) the greatest amount (d) the ionic form

22. The concentration unit of ***molarity*** (***M***) is correctly expressed in which of the following units?
(a) mol/kg (b) mol/L (c) gl/L (d) kg/mol

23. Which of the following is a spectator ion in the reaction (hint: compare complete ionic to net ionic):
$NaCl\,(aq) + AgNO_3\,(aq) \rightarrow AgCl\,(s) + NaNO_3\,(aq)$?
(a) Na^+ (b) Ag^+ (c) Cl^- (d) AgCl

24. Which statement t best describes the principle of "like dissolves like"?
(a) polar solutes dissolve in polar solvents (b) nonpolar solutes dissolve in nonpolar solvents
(c) ionic solutes dissolve in polar solvents (d) all of choices (a) to (c)

25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
(a) ***(s)*** (b) ***(l)*** (c) ***(g)*** (d) ***(aq)***

Circle either TRUE or FALSE whether the statement is true or false.

26. A solution that conducts electricity must contain ions.
True False

27. The reaction HCl (*aq*) + NaOH (*aq*) → H_2O (*l*) + NaCl (*aq*) is an acid-base neutralization reaction type.
True False

28. The balancing of a chemical reaction equation involves changing the subscripts of a chemical compound formula, like changing **$CuCl_2$** to **CuCl**
True False

29. The oxidation number of a pure element is always zero.
True False

30. In a chemical reaction equation, compounds to the ***right*** of the **reaction arrow** (yield symbol) are called ***reactants***.
True False

31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.
True False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.
True False

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.
True False

34. At STP, the temperature is 0°C and pressure is 1 atm.
True False

35. The symbol ***n*** in the Ideal Gas Law equation (*PV=nRT*) is known as the ***universal gas constant***.
True False

Chemistry 3A – Spring 2026 **Grecia Pereira Pedraza** **0667404** RETAKE VERSION

Sections 19320-19321

Midterm Examination #3 8 May 2026

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11. Which of the following is the correct oxidation number of ***hydrogen*** in H_2O molecule?
(a) 0 (b) -1 (c) +1 (d) +2

12. A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
(a) non-electrolyte (b) weak electrolyte (c) strong electrolyte (d) nonpolar molecule

13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2?
(a) -1 (b) 0 (c) +1 (d) +7

14. The strongest type of ***intermolecular force*** that exists between two molecules of H_2O (water) is:
(a) dispersion forces (London) (b) dipole-dipole forces
(c) ion-dipole forces (d) hydrogen bonding

15. How many grams of KOH are required to prepare 1.00 L of a 0.500 M solution?
(molar mass of KOH = 56.11 g/mol)
(a) 5.61 g (b) 28.1 g (c) 56.1 g (d) 112.2 g

16. Which is a TRUE statement of the **Kinetic Molecular Theory** for an *ideal gas*?
(a) gas molecules exert significant attractive forces on each other
(b) the molecules that make up the gas phase must interact with their liquid phase
(c) gas collisions are completely elastic (energy is not lost in collision)
(d) the volume (*V*) of the gas is inversely proportional to the number of moles (*n*)

17. Which of the following compounds is ***insoluble*** (forms precipitate) in water?
(a) KNO_3 (b) Na_2SO_4 (c) AgCl (d) LiCl

18. A solution that has not yet reached its upper limit of dissolved solute is described as:
(a) unsaturated (b) supersaturated (c) saturated (d) concentrated

19. A gas has initial pressure of 2.00 atm and volume of 5.00 L. If the volume is compressed to 2.50 L at constant temperature, what is the new pressure?
(a) 1.00 atm (b) 4.00 atm (c) 5.00 atm (d) 10.0 atm

20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
(a) S, Cl, Br (b) Cl, Br, I (c) N, O, F (d) Na, K, Li

21. In a sugar-water solution, the sugar is the ***solute*** and the water is the ***solvent***. The solvent is defined as the component present in:
(a) the smallest amount (b) the solid phase
(c) the greatest amount (d) the ionic form

22. The concentration unit of ***molarity*** (***M***) is correctly expressed in which of the following units?
(a) mol/kg (b) mol/L (c) gl/L (d) kg/mol

23. Which of the following is a spectator ion in the reaction (hint: compare complete ionic to net ionic):
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24. Which statement t best describes the principle of "like dissolves like"?
(a) polar solutes dissolve in polar solvents (b) nonpolar solutes dissolve in nonpolar solvents
(c) ionic solutes dissolve in polar solvents (d) all of choices (a) to (c)

25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
(a) **(*s*)** (b) **(*l*)** (c) **(*g*)** (d) **(*aq*)**

Circle either TRUE or FALSE whether the statement is true or false.

26. A solution that conducts electricity must contain ions.
True False

27. The reaction HCl (*aq*) + NaOH (*aq*) → H_2O (*l*) + NaCl (*aq*) is an acid-base neutralization reaction type.
True False

28. The balancing of a chemical reaction equation involves changing the subscripts of a chemical compound formula, like changing **$CuCl_2$** to **CuCl**
True False

29. The oxidation number of a pure element is always zero.
True False

30. In a chemical reaction equation, compounds to the ***right*** of the **reaction arrow** (yield symbol) are called ***reactants***.
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31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.
True False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.
True False

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.
True False

34. At STP, the temperature is 0°C and pressure is 1 atm.
True False

35. The symbol ***n*** in the Ideal Gas Law equation (*PV=nRT*) is known as the ***universal gas constant***.
True False

Chemistry 3A – Spring 2026 **Xionaya Ocampo** **0972874** RETAKE VERSION
Sections 19320-19321
Midterm Examination #3 8 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law

2. Convert a pressure of 1.50 atm to millimeters of mercury (mmHg)
(a) 507 mmHg (b) 760 mmHg (c) 1,140 mmHg (d) 1,500 mmHg

3. In a redox reaction, the species that ***loses electrons*** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) unchanged

4. A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., A + B → AB) is what type of reaction?
(a) decomposition (b) single displacement (SD)
(c) double displacement (DD) (d) combination (synthesis) reaction

5. What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M

6. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M

7. A solution is prepared by mixing 25.0 g of solute with 75.0 g of solvent. What is the percent by mass (m/m%) of the solute?
(a) 25.0% (b) 33.3% (c) 75.0% (d) 100.0%

8. Calculate the **pressure** of n = 2.00 mol of gas at T = 273 K, V = 44.8 L. (R = 0.0821 L·atm/[mol·K])
(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm

9. Convert a **pressure** of 760 mmHg to atmospheres (atm).
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15. How many grams of KOH are required to prepare 1.00 L of a 0.500 M solution?
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(a) ***(s)*** (b) ***(l)*** (c) ***(g)*** (d) ***(aq)***

Circle either TRUE or FALSE whether the statement is true or false.

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True False

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29. The oxidation number of a pure element is always zero.
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True False

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True False

35. The symbol ***n*** in the Ideal Gas Law equation (*PV=nRT*) is known as the ***universal gas constant***.
True False

Chemistry 3A – Spring 2026 **Steffany Joy Chan Sinsuan** **0973521** RETAKE VERSION

Sections 19320-19321

Midterm Examination #3 8 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law

2. Convert a pressure of 1.50 atm to millimeters of mercury (mmHg)
(a) 507 mmHg (b) 760 mmHg (c) 1,140 mmHg (d) 1,500 mmHg

3. In a redox reaction, the species that ***loses electrons*** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) unchanged

4. A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., A + B → AB) is what type of reaction?
(a) decomposition (b) single displacement (SD)
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5. What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M

6. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M

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(a) ***(s)*** (b) ***(l)*** (c) ***(g)*** (d) ***(aq)***

Circle either TRUE or FALSE whether the statement is true or false.

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True False

Alexis Romero

Chemistry 3A – Spring 2026 **Alexis Romero** **0932310** RETAKE VERSION
Sections 19320-19321
Midterm Examination #3 8 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

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5. What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M
$M = \frac{0.50}{500.0\,mL}$

6. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
$C_1V_1 = C_2V_2$ $(2.5M \cdot 10mL) = (50.0mL \cdot C_2)$ $\frac{25\,M = C_2 \cdot 50.0mL}{50mL}$ $C_2 = \frac{50.0mL}{25}$
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M

7. A solution is prepared by mixing 25.0 g of solute with 75.0 g of solvent. What is the percent by mass (m/m%) of the solute?
$\frac{25.0}{75.0} \times 100 = \%$ $C_2 = \frac{25}{50.0mL}$
(a) 25.0% (b) 33.3% (c) 75.0% (d) 100.0%

Ideal gas law

8. Calculate the **pressure** of n = 2.00 mol of gas at T = 273 K, V = 44.8 L. (R = 0.0821 L·atm/[mol·K])
(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm
$P = nRT$ $2.00mol \times (0.0821 \cdot \frac{atm}{mol \cdot K}) \times 273K$ $\frac{}{44.8L}$
$1.000 \frac{593}{5} atm$

9. Convert a **pressure** of 760 mmHg to atmospheres (atm).
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Chemistry 3A – Spring 2026 **Emily Pastrana** **1049326** RETAKE VERSION

Sections 19320-19321

Midterm Examination #3

8 May 2026

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   (a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M

7. A solution is prepared by mixing 25.0 g of solute with 75.0 g of solvent. What is the percent by mass (m/m%) of the solute?
   (a) 25.0% (b) 33.3% (c) 75.0% (d) 100.0%

8. Calculate the **pressure** of n = 2.00 mol of gas at T = 273 K, V = 44.8 L. (R = 0.0821 L·atm/[mol·K])
   (a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm

9. Convert a **pressure** of 760 mmHg to atmospheres (atm).
   (a) 0.500 atm (b) 1.00 atm (c) 7.60 atm (d) 76.0 atm

10. Which of the following is a ***combustion*** reaction?
    (a) $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ (b) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
    (c) $HCl + NaOH \rightarrow NaCl + H_2O$ (d) $CaCO_3 \rightarrow CaO + CO_2$

11. Which of the following is the correct oxidation number of ***hydrogen*** in H_2O molecule?
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12. A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
    (a) non-electrolyte (b) weak electrolyte (c) strong electrolyte (d) nonpolar molecule

13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2?
    (a) -1 (b) 0 (c) +1 (d) +7

14. The strongest type of ***intermolecular force*** that exists between two molecules of H_2O (water) is:
(a) dispersion forces (London) (b) dipole-dipole forces
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15. How many grams of KOH are required to prepare 1.00 L of a 0.500 M solution?
(molar mass of KOH = 56.11 g/mol)
(a) 5.61 g (b) 28.1 g (c) 56.1 g (d) 112.2 g

16. Which is a TRUE statement of the **Kinetic Molecular Theory** for an *ideal gas*?
(a) gas molecules exert significant attractive forces on each other
(b) the molecules that make up the gas phase must interact with their liquid phase
(c) gas collisions are completely elastic (energy is not lost in collision)
(d) the volume (*V*) of the gas is inversely proportional to the number of moles (*n*)

17. Which of the following compounds is ***insoluble*** (forms precipitate) in water?
(a) KNO_3 (b) Na_2SO_4 (c) AgCl (d) LiCl

18. A solution that has not yet reached its upper limit of dissolved solute is described as:
(a) unsaturated (b) supersaturated (c) saturated (d) concentrated

19. A gas has initial pressure of 2.00 atm and volume of 5.00 L. If the volume is compressed to 2.50 L at constant temperature, what is the new pressure?
(a) 1.00 atm (b) 4.00 atm (c) 5.00 atm (d) 10.0 atm

20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
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(c) the greatest amount (d) the ionic form

22. The concentration unit of ***molarity*** (***M***) is correctly expressed in which of the following units?
(a) mol/kg (b) mol/L (c) gl/L (d) kg/mol

23. Which of the following is a spectator ion in the reaction (hint: compare complete ionic to net ionic):
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24. Which statement t best describes the principle of "like dissolves like"?
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25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
(a) **(*s*)** (b) **(*l*)** (c) **(*g*)** (d) **(*aq*)**

Circle either TRUE or FALSE whether the statement is true or false.

26. A solution that conducts electricity must contain ions.
True False

27. The reaction HCl (*aq*) + NaOH (*aq*) → H_2O (*l*) + NaCl (*aq*) is an acid-base neutralization reaction type.
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28. The balancing of a chemical reaction equation involves changing the subscripts of a chemical compound formula, like changing **$CuCl_2$** to **CuCl**
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29. The oxidation number of a pure element is always zero.
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33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.
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34. At STP, the temperature is 0°C and pressure is 1 atm.
True False

35. The symbol ***n*** in the Ideal Gas Law equation (*PV=nRT*) is known as the ***universal gas constant***.
True False

n = mol

Chemistry 3A – Spring 2026 **Noe Moises Carrillo** **0959275** RETAKE VERSION
Sections 19320-19321
Midterm Examination #3 8 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law

2. Convert a pressure of 1.50 atm to millimeters of mercury (mmHg)
(a) 507 mmHg (b) 760 mmHg (c) 1,140 mmHg (d) 1,500 mmHg

3. In a redox reaction, the species that ***loses electrons*** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) unchanged

4. A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., A + B → AB) is what type of reaction?
(a) decomposition (b) single displacement (SD)
(c) double displacement (DD) (d) combination (synthesis) reaction

5. What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M

6. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
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Circle either TRUE or FALSE whether the statement is true or false.

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True False

Chemistry 3A – Spring 2026 **Evelyn Garcia Hernandez** **0951887** RETAKE VERSION

Sections 19320-19321

Midterm Examination #3 8 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

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Mariam Smith
5-5-26
0483202

Chemistry 3A – Spring 2026
Sections 19320-19321
Midterm Examination #3

RETAKE VERSION

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

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9

7/10 T/F

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Chemistry 3A – Spring 2026 **Mitxell Mendez 1029129** RETAKE VERSION

Sections 19320-19321

Midterm Examination #1 7 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which element belongs to the **alkali metals** group?
(a) calcium (b) sodium (c) magnesium (d) chlorine

2. Which subatomic particle has a mass of approximately **1 amu** and a positive charge?
(a) electron (b) proton (c) neutron (d) none of the above

3. Which of the following is a **polyatomic ion**?
(a) Cl^- (b) Na^+ (c) SO_4^{2-} (d) O^{2-}

4. Convert 25.0 °C into Kelvin.
(a) 0 K (b) 25 K (c) 273 K (d) 298 K

5. A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density?
(a) 0.200 g/mL (b) 5.00 g/mL (c) 31.3 g/mL (d) 0.050 g/mL

6. Which of the following is the correct name for $Ca(NO_3)_2$?
(a) cobalt fluoride (b) calcium nitroxide (c) calcium nitrate (d) calcium oxide

7. Which of the following terms describes the distance between two wave crests?
(a) wave speed (b) frequency (c) wavelength (d) amplitude

8. Which of the following is the correct atomic notation for carbon-14?
(a) ${}^{12}_{6}C$ (b) ${}^{14}_{6}C$ (c) ${}^{6}_{14}C$ (d) ${}^{60}_{27}Co$

9. Convert 4.56×10^{-3} into standard notation
(a) 0.00456 (b) 4560 (c) 0.000456 (d) 4,560

10. Which of the following is the electron configuration for hydrogen?
(a) $1s^1$ (b) $1s^2$ (c) $1s^2 2s^1$ (d) $1s^2 2p^1$

11. Which of the following is TRUE about element bromine (Br)?
(a) It is a member of the *d*-block of elements
(b) It is a Group 1 element
(c) It is a non-metal
(d) none of the above

12. What term describes an electron normally in $n = 1$ shell acquiring energy to be in $n=2$ shell?
(a) emission (b) excited state (d) ground state (d) absorption

13. Express 0.0000721 in scientific notation with correct significant figures.
(a) 7.21×10^{-4} (b) 7.21×10^{-5} (c) 7.21×10^{-6} (d) 7.21×10^{-7}

14. When you complete the operation on the expression 1.450 × 3.01 / 10.025, how many significant digits will the final result have? ('/' = division operation)
(a) 2 (b) 3 (c) 4 (d) 5

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?
(a) law (b) hypothesis (c) theory (d) experimentation

16. Which of these is a polyatomic ion?
(a) Ne (b) CO_3^{2-} (c) Br^- (d) Na^+

17. Which of these is the ground-state electron configuration for the **carbon (C)** atom?
(a) $1s^2 2s^2 2p^6$ (b) $1s^2 2s^2 2p^2$ (c) $1s^2 2s^2 2p^4$ (d) $1s^2 2s^2$

18. How many quantum numbers are required to uniquely define the state of an electron in an atom?
(a) 1 (b) 2 (c) 3 (d) 4

19. Perform the operation: 3.456 + 12.1 + 0.0045. Report with correct decimal places.
(a) 15.5605 (b) 15.56 (c) 15.6 (d) 15.561

20. The label on the bottle has the formula $FeCl_3$. What is its name using the Stock system?
(a) iron chloride (b) iron(II) chloride (c) iron(III) chloride (d) iron monochloride

21. One of the quantum numbers of an electron is related to the shape of the electron's orbital and is represented by a letter in this set: {*s*, *p*, *d*, *f*}. Which quantum number is this?
(a) *n* (b) *l* (c) m_s (d) Z

22. Which of these subatomic particles is **NOT** found in the nucleus of the atom?
(a) proton (b) neutron (c) electron (d) none of the choices (a) to (c)

23. What is a TRUE statement regarding an element in the Periodic Table?
(a) the number of neutrons and protons for every element must be equal
(b) the atomic number (Z value) indicates the number of protons it has
(c) the mass number (A value) indicates the number of electrons and protons it has
(d) the isotope of the element is determined by the number of electrons

24. What are the Group 2 elements called?
(a) Noble gases (b) Alkali metals (c) Alkaline Earth metals (d) Halogens

25. An isotope of an element has atomic number (Z) = 15 and mass number (A) of 31. How many neutrons are in the nucleus?
(a) 15 (b) 16 (c) 31 (d) 46

Circle either TRUE or FALSE whether the statement is true or false.

26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.
True False

27. Ionic compounds with monatomic elements are typically formed between metals and nonmetals.
True False

28. The speed/velocity of a electromagnetic wave can be calculated by dividing the frequency by the wavelength: $c = \nu / \lambda$
True False

29. The **p-block** of elements in the Periodic Table starts in Period 1.
True False

30. The **Pauli Exclusion Principle** states that an orbital can have only two electrons which are paired up with opposite spins
True False

31. According to the rules for significant figures, **conversion factors** (like 1 m = 100 cm) are considered to have infinite significant digits and do not affect the precision of a calculation.
True False

32. The atomic number (*Z*) represents the number of protons in an atom.
True False

33. The only subatomic particles allowed in the nucleus of the atom are **protons**.
True False

34. Noble gases have complete valence shells, which is why they do not form compounds with other elements.
True False

35. The law of conservation of mass states that matter **CAN** be created or destroyed in chemical reactions.
True False

Chemistry 3A – Spring 2026 **Shantal Escareno 1047747** RETAKE VERSION

Sections 19320-19321

Midterm Examination #1 7 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which element belongs to the **alkali metals** group?
(a) calcium (b) sodium (c) magnesium (d) chlorine

2. Which subatomic particle has a mass of approximately **1 amu** and a positive charge?
(a) electron (b) proton (c) neutron (d) none of the above

3. Which of the following is a **polyatomic ion**?
(a) Cl^- (b) Na^+ (c) SO_4^{2-} (d) O^{2-}

4. Convert 25.0 °C into Kelvin.
(a) 0 K (b) 25 K (c) 273 K (d) 298 K

5. A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density?
(a) 0.200 g/mL (b) 5.00 g/mL (c) 31.3 g/mL (d) 0.050 g/mL

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True False

Chemistry 3A – Spring 2026 **Brandon William Hargrave** **0894716** RETAKE VERSION

Sections 19320-19321

Midterm Examination #3 7 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law

2. Convert a pressure of 1.50 atm to millimeters of mercury (mmHg)
(a) 507 mmHg (b) 760 mmHg (c) 1,140 mmHg (d) 1,500 mmHg

3. In a redox reaction, the species that ***loses electrons*** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) unchanged

4. A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., A + B → AB) is what type of reaction?
(a) decomposition (b) single displacement (SD)
(c) double displacement (DD) (d) combination (synthesis) reaction

5. What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M

6. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M

7. A solution is prepared by mixing 25.0 g of solute with 75.0 g of solvent. What is the percent by mass (m/m%) of the solute?
(a) 25.0% (b) 33.3% (c) 75.0% (d) 100.0%

8. Calculate the **pressure** of n = 2.00 mol of gas at T = 273 K, V = 44.8 L. (R = 0.0821 L·atm/[mol·K])
(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm

9. Convert a **pressure** of 760 mmHg to atmospheres (atm).
(a) 0.500 atm (b) 1.00 atm (c) 7.60 atm (d) 76.0 atm

10. Which of the following is a ***combustion*** reaction?
(a) $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ (b) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(c) $HCl + NaOH \rightarrow NaCl + H_2O$ (d) $CaCO_3 \rightarrow CaO + CO_2$

11. Which of the following is the correct oxidation number of ***hydrogen*** in H_2O molecule?
(a) 0 (b) -1 (c) +1 (d) +2

12. A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
(a) non-electrolyte (b) weak electrolyte (c) strong electrolyte (d) nonpolar molecule

13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2?
(a) -1 (b) 0 (c) +1 (d) +7

14. The strongest type of ***intermolecular force*** that exists between two molecules of H_2O (water) is:
(a) dispersion forces (London) (b) dipole-dipole forces
(c) ion-dipole forces (d) hydrogen bonding

15. How many grams of KOH are required to prepare 1.00 L of a 0.500 M solution? (molar mass of KOH = 56.11 g/mol)
(a) 5.61 g (b) 28.1 g (c) 56.1 g (d) 112.2 g

16. Which is a TRUE statement of the **Kinetic Molecular Theory** for an *ideal gas*?
(a) gas molecules exert significant attractive forces on each other
(b) the molecules that make up the gas phase must interact with their liquid phase
(c) gas collisions are completely elastic (energy is not lost in collision)
(d) the volume (*V*) of the gas is inversely proportional to the number of moles (*n*)

17. Which of the following compounds is ***insoluble*** (forms precipitate) in water?
(a) KNO_3 (b) Na_2SO_4 (c) AgCl (d) LiCl

18. A solution that has not yet reached its upper limit of dissolved solute is described as:
(a) unsaturated (b) supersaturated (c) saturated (d) concentrated

19. A gas has initial pressure of 2.00 atm and volume of 5.00 L. If the volume is compressed to 2.50 L at constant temperature, what is the new pressure?
(a) 1.00 atm (b) 4.00 atm (c) 5.00 atm (d) 10.0 atm

20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
(a) S, Cl, Br (b) Cl, Br, I (c) N, O, F (d) Na, K, Li

21. In a sugar-water solution, the sugar is the ***solute*** and the water is the ***solvent***. The solvent is defined as the component present in:
(a) the smallest amount (b) the solid phase
(c) the greatest amount (d) the ionic form

22. The concentration unit of ***molarity*** (***M***) is correctly expressed in which of the following units?
(a) mol/kg (b) mol/L (c) gl/L (d) kg/mol

23. Which of the following is a spectator ion in the reaction (hint: compare complete ionic to net ionic):
$NaCl\ (aq) + AgNO_3\ (aq) \rightarrow AgCl\ (s) + NaNO_3\ (aq)$?
(a) Na^+ (b) Ag^+ (c) Cl^- (d) AgCl

24. Which statement t best describes the principle of "like dissolves like"?
(a) polar solutes dissolve in polar solvents (b) nonpolar solutes dissolve in nonpolar solvents
(c) ionic solutes dissolve in polar solvents (d) all of choices (a) to (c)

25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
(a) ***(s)*** (b) ***(l)*** (c) ***(g)*** (d) ***(aq)***

Circle either TRUE or FALSE whether the statement is true or false.

26. A solution that conducts electricity must contain ions.
True False

27. The reaction HCl (*aq*) + NaOH (*aq*) → H_2O (*l*) + NaCl (*aq*) is an acid-base neutralization reaction type.
True False

28. The balancing of a chemical reaction equation involves changing the subscripts of a chemical compound formula, like changing **$CuCl_2$** to **CuCl**
True False

29. The oxidation number of a pure element is always zero.
True False

30. In a chemical reaction equation, compounds to the ***right*** of the **reaction arrow** (yield symbol) are called ***reactants***.
True False

31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.
True False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.
True False

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.
True False

34. At STP, the temperature is 0°C and pressure is 1 atm.
True False

35. The symbol ***n*** in the Ideal Gas Law equation (*PV=nRT*) is known as the ***universal gas constant***.
True False